

Natural Language Processing Notes

Turning text into patterns you can inspect

Text is structured information

Natural language processing explores how software can work with human language. Text contains meaning, but it also contains observable patterns: words, punctuation, repeated phrases, and relationships between sentences. A practical analysis starts by deciding which patterns will help answer a question. A document dashboard can reveal frequent terms without claiming to understand every idea in the document.

The first step is often tokenization, which divides text into units. Word tokens can be useful for frequency statistics, while sentence tokens support readability estimates. Different tokenization rules handle punctuation, contractions, and compound terms differently. Write down the rule so that users understand what a count represents.

Normalization and stopwords

Normalization makes comparisons more consistent. Case folding treats Learning and learning as the same word for many frequency tasks. Whitespace cleanup helps readability, while preserving paragraph boundaries retains some document structure. Removing punctuation may help one analysis but discard useful information for another. The appropriate choices depend on the question.

Stopwords are frequent function words such as the, and, or of. Filtering them can make a keyword list easier to scan. A stopword list is a useful heuristic rather than a universal definition of meaningful language. Domain-specific terms may also need filtering. Always keep the original extracted text available so that users can inspect the evidence.

Frequency and lexical diversity

A frequency table counts how often each token occurs. Sorting that table reveals repeated terms and supports a simple bar chart. Frequency can point to a topic, but repetition does not prove importance. Boilerplate, repeated headers, and extraction artifacts may appear often too. Reviewing the document alongside the chart keeps the analysis grounded.

Lexical diversity compares the number of unique words with the total word count. A short document can have a higher ratio than a long document even when both use varied language. Use the ratio as a descriptive statistic and avoid treating it as a quality score. Similarly, reading time is an estimate based on an assumed reading speed.

Going beyond counts

Search connects a statistical overview to specific passages. Literal case-insensitive search is predictable and easy to explain. Snippets show the context surrounding each match, while highlighted page text helps users locate the full passage. Search results must remain safely escaped when displayed in a web interface.

More advanced analysis can include named entities, topics, summaries, and language-specific tokenization. Each approach introduces assumptions and evaluation requirements. Begin with transparent statistics, retain context, and clearly describe limitations. A trustworthy text tool helps users inspect its results instead of asking them to accept unexplained conclusions.